

# 2026 March Int I Module 1

## Question 1

If  $6 + x = 70$ , what is the value of  $48 + 8x$ ?

## Question 2

| Type of entree | Number of people |
|----------------|------------------|
| Beef           | 17               |
| Chicken        | 20               |
| Fish           | 6                |
| Vegetarian     | 7                |
| <b>Total</b>   | <b>50</b>        |

A total of 50 people attended a conference and were offered 4 types of entrees. The table shows the number of people who chose each type of entree. If one of these people is selected at random, what is the probability of selecting a person who chose a vegetarian entree?

- A)  $7/100$
- B)  $7/50$
- C)  $1/4$
- D)  $7/10$

## Question 3

$$y = 30x + 100$$

The equation gives the total cost  $y$ , in dollars, for a small business to produce  $x$  shirts. How many shirts can the business produce for a total cost of \$550?

- A)  $-3$
- B) 15
- C) 18
- D) 21

## Question 4

Triangles  $XYZ$  and  $RST$  are congruent, where  $XY = 654$  and  $YZ = 654$ . If  $RS = ST$ , what is the length of segment  $RS$ ?

- A) 327
- B) 654
- C) 1,308
- D) 1,962

## Question 5

A length of 590 meters is equal to how many decimeters? (1 meter = 10 decimeters)

## Question 6

An object traveled 0.20 meters for every 4.0 seconds it was in motion. If the object was in motion for 24 seconds, what is the total distance, in meters, the object traveled?

## Question 7

The function  $f$  is defined by  $f(x) = 6(4)^x$ . What is the value of  $f(x)$  when  $x = 3$ ?

- A) 64
- B) 81
- C) 258
- D) 384

## Question 8

The number of cells of the bacteria *Bacillus megaterium* doubles every 25 minutes during its growth period when placed in a medium of sucrose-salts in a controlled laboratory environment. If a sample of 980 cells of *Bacillus megaterium* is placed in sucrose-salts in a controlled laboratory environment, which function  $f$  best represents the number of cells of the bacteria  $x$  minutes after the growth period begins, where  $x$  is less than the number of minutes in the growth period?

- A)  $f(x) = 980\left(\frac{1}{2}\right)^{\frac{x}{25}}$
- B)  $f(x) = 980(2)^{\frac{x}{25}}$
- C)  $f(x) = 980\left(\frac{1}{3}\right)^{\frac{x}{25}}$
- D)  $f(x) = 980(3)^{\frac{x}{25}}$

## Question 9

A model predicts that a certain elephant calf weighed 208 pounds when it was born and that the calf gained 3 pounds per day in its first year of life. This model is defined by an equation in the form  $f(x) = a + bx$ , where  $f(x)$  is the predicted weight, in pounds, of the elephant calf  $x$  days after it was born, and  $a$  and  $b$  are constants. What is the value of  $a$ ?

**Question 10**

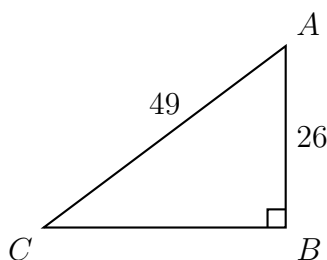
The expression  $13x^3 + 19x^3 - 17x^3$  is equivalent to  $bx^3$ , where  $b$  is a constant. What is the value of  $b$ ?

**Question 11**

$$x^2 - 20x = 0$$

Which of the following is a solution to the given equation?

- A) 40
- B) 20
- C) 10
- D)  $\sqrt{20}$

**Question 12**

*Note: Figure not drawn to scale.*

In the right triangle shown, what is the value of  $\cos A$ ?

- A)  $1/49$
- B)  $1/26$
- C)  $26/49$
- D)  $49/26$

**Question 13**

| $x$ | $t(x)$         |
|-----|----------------|
| 29  | 15             |
| 32  | $\frac{33}{2}$ |
| 51  | 26             |

For the linear function  $t$ , the table shows three values of  $x$  and their corresponding values of  $t(x)$ . Function  $v$  is defined by  $v(x) = t(x) - 4$ . What is the  $x$ -intercept of the graph of  $y = v(x)$  in the  $xy$ -plane?

- A)  $(-1, 0)$
- B)  $(3, 0)$
- C)  $(7, 0)$
- D)  $(8, 0)$

### Question 14

For the polynomial function  $f$ , the graph of  $y = f(x)$  in the  $xy$ -plane passes through the points  $(-5, 0)$ ,  $(3, 0)$ , and  $(4, 0)$ . Which of the following must be a factor of  $f(x)$ ?

- A)  $x + 3$
- B)  $x + 4$
- C)  $x - 3$
- D)  $x - 5$

### Question 15

What is the  $x$ -intercept of a line in the  $xy$ -plane that has a slope of  $-4/7$  and passes through the point  $(0, 12)$ ?

- A)  $(-21, 0)$
- B)  $(-4, 0)$
- C)  $(7, 0)$
- D)  $(21, 0)$

### Question 16

A circle has a circumference of  $10\pi$  centimeters. What is the diameter, in centimeters, of the circle?

### Question 17

A neighborhood consists of a 3-hectare park and a 34-hectare residential area. The total number of trees in the neighborhood is 4,832. The equation  $3x + 34y = 4,832$  represents this situation. Which of the following is the best interpretation of  $x$  in this context?

- A) The average number of trees per hectare in the park
- B) The average number of trees per hectare in the residential area
- C) The total number of trees in the park
- D) The total number of trees in the residential area

## Question 18

In the  $xy$ -plane, an equation of circle  $A$  is  $(x - 5)^2 + (y - 7)^2 = 16$ . Circle  $B$  has the same center as circle  $A$  but has a radius that is twice the radius of circle  $A$ . Which equation represents circle  $B$ ?

- A)  $(x - 5)^2 + (y - 7)^2 = 32$
- B)  $(x - 5)^2 + (y - 7)^2 = 64$
- C)  $(x - 5)^2 + (y - 7)^2 = 128$
- D)  $(x - 5)^2 + (y - 7)^2 = 256$

## Question 19

| Number of employees | Number of stores |
|---------------------|------------------|
| 1 to 6              | 2                |
| 7 to 12             | 3                |
| 13 to 18            | 1                |
| 19 to 24            | 6                |
| 25 to 30            | 1                |

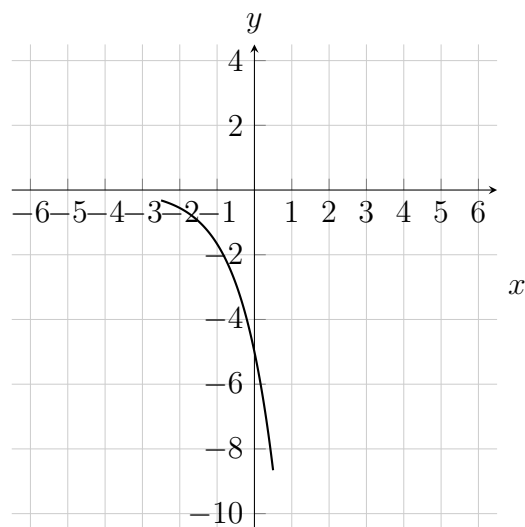
The table shown summarizes the number of employees at each of the 13 stores in a shopping plaza. Which of the following could be the median number of employees for the stores in the shopping plaza?

- A) 1
- B) 9
- C) 15
- D) 21

## Question 20

For the positive quantities  $m$ ,  $q$ , and  $r$ ,  $m$  is 20% of  $m + 9 + r$ ,  $q$  is 30% of  $q + r$ , and the value of  $r$  is 1,358. What is the value of  $m$ ?

## Question 21



The graph of  $y = f(x) + 2$  is shown, where  $f$  is defined by an equation of the form  $f(x) = a(b)^x + c$ . If  $a$ ,  $b$ , and  $c$  are integer constants and  $c \leq 0$ , which equation could define  $f$ ?

- A)  $f(x) = -5(3)^x$
- B)  $f(x) = -5(3)^x - 1$
- C)  $f(x) = -5(3)^x - 3$
- D)  $f(x) = -5(3)^x - 5$

## Question 22

$$30x = 1,800y - 2,700$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

- A)  $\frac{1}{30}x = 2y$
- B)  $\frac{1}{30}x = 60y - 90$
- C)  $x = 2y$
- D)  $x = 60y - 90$

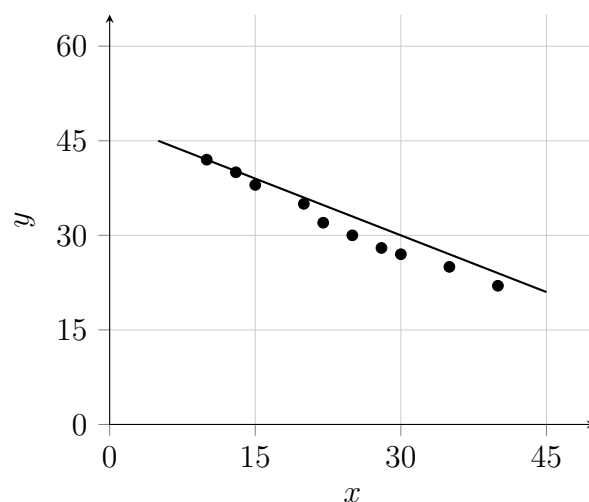
## Math Module 2

## Question 1

A candle is made of 18 ounces of wax. When the candle is burning, the amount of wax in the candle decreases by 1 ounce every 4 hours. If 8 ounces of wax remain in this candle, for how many hours has it been burning?

- A) 3
- B) 7
- C) 32
- D) 40

## Question 2



The scatterplot shows the relationship between two variables,  $x$  and  $y$ , for data set  $A$ . A line of best fit for the data is also shown. Data set  $B$  is created by subtracting 8 units from the value of  $y$  for each data point from data set  $A$ . Which of the following is closest to the  $y$ -coordinate of the  $y$ -intercept of the line of best fit for data set  $B$ ?

- A) 39.17
- B) 48
- C) 52.22
- D) 72.33

## Question 3

$$q(x) = 12(2^x)$$

Which table gives three values of  $x$  and their corresponding values of  $q(x)$  for function  $q$ ?

- A) 

|        |     |   |    |
|--------|-----|---|----|
| $x$    | -1  | 0 | 1  |
| $q(x)$ | -24 | 0 | 24 |

B) 

|        |               |     |      |
|--------|---------------|-----|------|
| $x$    | $-1$          | $0$ | $1$  |
| $q(x)$ | $\frac{1}{6}$ | $2$ | $24$ |

C) 

|        |               |      |      |
|--------|---------------|------|------|
| $x$    | $-1$          | $0$  | $1$  |
| $q(x)$ | $\frac{1}{6}$ | $12$ | $24$ |

D) 

|        |      |      |      |
|--------|------|------|------|
| $x$    | $-1$ | $0$  | $1$  |
| $q(x)$ | $6$  | $12$ | $24$ |

### Question 4

For the exponential function  $g$ , the value of  $g(x)$  doubles for every increase of 2 in the value of  $x$ . If  $g(5) = 13$ , what is the value of  $g(7)$ ?

### Question 5

The length of a rectangle's diagonal is  $\sqrt{34}$ , and the length of one of the rectangle's sides is 5. What is the perimeter of the rectangle?

- A) 34
- B) 16
- C) 15
- D) 8

### Question 6

$$2x + 12y = 6$$

$$x + 8y = 1$$

The solution to the given system of equations is  $(x, y)$ . What is the value of  $x + 4y$ ?

- A) 9
- B) 7
- C) 5
- D) 1

## Question 7

The expression  $\frac{6x^{24}}{18x^7}$  is equivalent to  $\frac{1}{3}x^b$ , where  $b$  is a constant and  $x > 0$ . What is the value of  $b$ ?

- A)  $7/24$
- B)  $24/7$
- C) 17
- D) 31

## Question 8

An object is kicked from a platform. The equation  $h = -4.9t^2 + 8t + 11$  represents this situation, where  $h$  is the height of the object above the ground, in meters,  $t$  seconds after it is kicked. According to the equation, what is the height, in meters, from which the object was kicked?

## Question 9

| Number of cars | Maximum number of passengers and crew |
|----------------|---------------------------------------|
| 3              | 139                                   |
| 6              | 271                                   |
| 10             | 447                                   |

The table shows the linear relationship between the number of cars,  $c$ , on a commuter train and the maximum number of passengers and crew,  $p$ , that the train can carry. Which equation represents the linear relationship between  $c$  and  $p$ ?

- A)  $44c - p = -7$
- B)  $44c - p = 7$
- C)  $44p - c = -7$
- D)  $44p - c = 7$

## Question 10

$$x(x + 3) - 1 = 0$$

What are the solutions to the given equation?

- A)  $x = \frac{-3 \pm \sqrt{13}}{2}$
- B)  $x = \frac{-3 \pm \sqrt{5}}{2}$

C)  $x = \frac{3 \pm \sqrt{13}}{2}$

D)  $x = \frac{3 \pm \sqrt{5}}{2}$

### Question 11

$$y \leq x + 14$$

$$y \geq -3x - 14$$

Which point  $(x, y)$  is a solution to the given system of inequalities in the  $xy$ -plane?

A)  $(0, -15)$

B)  $(0, 15)$

C)  $(-8, 0)$

D)  $(8, 0)$

### Question 12

$$35x + 3 = k(7x + 3) + 7x$$

In the given equation,  $k$  is a constant. The equation has exactly one solution. Which value CANNOT be the value of  $k$ ?

A) 5

B) 4

C) 0

D)  $-4$

### Question 13

| $x$   | $y$  |
|-------|------|
| $-12$ | $46$ |
| $a$   | $16$ |
| $2$   | $b$  |

The table shows three values of  $x$  and their corresponding values of  $y$ , where  $a$  and  $b$  are constants. There is a linear relationship between  $x$  and  $y$ . In the  $xy$ -plane, the  $y$ -intercept of the line representing this relationship is  $(0, -14)$ . What is the value of  $a + b$ ?

A)  $-30$

B)  $-29$

C)  $-20$

D)  $-19$

### Question 14

Line  $s$  is perpendicular to line  $t$  in the  $xy$ -plane. Line  $s$  passes through the points  $(6, 7/8)$  and  $(-3, -17/8)$ . Line  $t$  passes through the points  $(-8, 39)$  and  $(k, 78)$ . What is the value of  $k$ ?

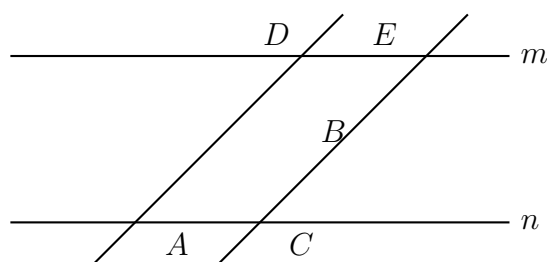
### Question 15

$$y = 2x^2 - 24x + 54$$

The given equation represents a parabola in the  $xy$ -plane. Which of the following equations that represent the same parabola displays the  $x$ -intercepts as constants or coefficients?

- A)  $y = 2(x^2 - 12x) + 54$
- B)  $y = 2x(x - 12) + 54$
- C)  $y = 2(x - 6)^2 - 18$
- D)  $y = 2(x - 9)(x - 3)$

### Question 16



*Note: Figure not drawn to scale.*

In the figure, line  $m$  is parallel to line  $n$ , and lines  $AE$  and  $CD$  intersect at point  $B$ . Which additional piece of information is sufficient to prove that triangle  $ABC$  is congruent to triangle  $EBD$ ?

- A)  $AB = 12$  and  $DB = 12$ .
- B)  $AB = 12$  and  $EB = 12$ .
- C) Triangles  $ABC$  and  $EBD$  are isosceles.
- D) No additional information is necessary to determine that the two triangles are congruent.

## Question 17

$$f(x) = 12(7)^x$$

The function  $f$  is defined by the given equation. If  $g(x) = f(x+2)$ , which of the following equations defines the function  $g$ ?

- A)  $g(x) = 24(7)^x$
- B)  $g(x) = 588(7)^x$
- C)  $g(x) = 24(14)^x$
- D)  $g(x) = 144(49)^x$

## Question 18

The value of a painting increased by 179% from the end of 2017 to the end of 2018 and then decreased by 23% from the end of 2018 to the end of 2019. What was the net percentage increase in the value of the painting from the end of 2017 to the end of 2019?

- A) 114.83%
- B) 137.83%
- C) 156.00%
- D) 243.17%

## Question 19

$$\frac{2}{7}(2x+7)\left(x+\sqrt{2k+7}\right)\left(x-\sqrt{2k+7}\right)=0$$

In the given equation,  $k$  is a positive constant. The product of the solutions to the equation is 63. What is the value of  $k$ ?

## Question 20

$$4x^2 - px + w = -87$$

In the given equation,  $p$  and  $w$  are integer constants. The equation has exactly one real solution. Which is **NOT** a possible value of  $w$ ?

- A)  $-23$
- B)  $13$
- C)  $81$
- D)  $313$

**Question 21**

|                  | Area (square inches) |
|------------------|----------------------|
| Trapezoid $ABCD$ | 440                  |
| Trapezoid $EFGH$ | 11,000               |

Trapezoid  $ABCD$  is similar to trapezoid  $EFGH$ , where  $A$ ,  $B$ ,  $C$ , and  $D$  correspond to  $E$ ,  $F$ ,  $G$ , and  $H$ , respectively. The length of side  $AB$  is 55 inches. What is the length, in inches, of side  $EF$ ?

**Question 22**

In triangle  $JKL$ , the measure of angle  $J$  is  $(90b)^\circ$ , the measure of angle  $K$  is  $(69a)^\circ$ , and the measure of angle  $L$  is  $(21a)^\circ$ , where  $a$  and  $b$  are constants. Which of the following must be true?

- A)  $\cos L > \sin K$
- B)  $\cos L = \sin K$
- C)  $\cos L < \sin K$
- D) There is not enough information to compare the values of  $\cos L$  and  $\sin K$

# Answer Sheet

## Module 1

|     |          |          |          |      |     |          |          |     |    |
|-----|----------|----------|----------|------|-----|----------|----------|-----|----|
| 1   | 2        | 3        | 4        | 5    | 6   | 7        | 8        | 9   | 10 |
| 560 | <i>B</i> | <i>B</i> | <i>B</i> | 5900 | 1.2 | <i>D</i> | <i>B</i> | 208 | 15 |

|          |          |          |          |          |    |          |          |          |     |
|----------|----------|----------|----------|----------|----|----------|----------|----------|-----|
| 11       | 12       | 13       | 14       | 15       | 16 | 17       | 18       | 19       | 20  |
| <i>B</i> | <i>C</i> | <i>C</i> | <i>C</i> | <i>D</i> | 10 | <i>A</i> | <i>B</i> | <i>D</i> | 485 |

|          |          |
|----------|----------|
| 21       | 22       |
| <i>A</i> | <i>A</i> |

## Module 2

|          |          |          |    |          |          |          |    |          |          |
|----------|----------|----------|----|----------|----------|----------|----|----------|----------|
| 1        | 2        | 3        | 4  | 5        | 6        | 7        | 8  | 9        | 10       |
| <i>D</i> | <i>A</i> | <i>D</i> | 26 | <i>B</i> | <i>C</i> | <i>C</i> | 11 | <i>A</i> | <i>A</i> |

|          |          |          |     |          |          |          |          |     |          |
|----------|----------|----------|-----|----------|----------|----------|----------|-----|----------|
| 11       | 12       | 13       | 14  | 15       | 16       | 17       | 18       | 19  | 20       |
| <i>D</i> | <i>B</i> | <i>A</i> | -21 | <i>D</i> | <i>B</i> | <i>B</i> | <i>A</i> | 5.5 | <i>C</i> |

|     |          |
|-----|----------|
| 21  | 22       |
| 275 | <i>D</i> |